

Multi-block PLS modeling for L-shape data structures with applications to mixture modeling

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Abstract

In some situations blocks of data are available from different sources. These data matrices often have no common dimension and are connected in a T- or L-shape structure. The example treated in this paper involves mixture modeling where blocks of data are available on the final product properties (**Y**), on the process conditions used to manufacture the products (**Z**), on the ratios of raw materials used in the formulations (**R**) and on the properties of each of those raw materials (**X**). In previous research [K. Muteki, J.F. MacGregor, T. Ueda, On the rapid development of new polymer blends: The optimal selection of materials and blend ratios, Industrial Engineering Chemical Research, Published on-line, May 25 (in press)] T- or L-shaped data matrices were combined using ideal or other known mixing rules to achieve a block data structure with a single common dimension as required in regression modeling. In this paper, we present an alternative multi-block PLS regression approach that does not require the use of such mixing rules to combine the data. Rather, it directly builds a multi-block PLS model for the T- or L-shaped structure. The approach is very flexible and provides increased interpretability on the relationships among the blocks. Furthermore, the method is extended to allow for nonlinear modeling between any of the adjacent data blocks, thereby allowing for non-ideal situations where ideal mixing rules do not hold or are unknown, and where the process is nonlinear. The approach is illustrated using data from an industrial coke making operation where many coals having different properties are blended and processed in coke ovens.

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1. Introduction

Rapidly developing new products required by customers with minimum costs is of paramount importance to the chemical industry. Muteki et al. [1] have presented a data-based modeling and optimization strategy that simultaneously takes into account the selection of raw materials, the selection of the ratios in which to blend them and the selection of the process conditions in order to achieve desired final product properties at a minimum cost. The mixture PLS models proposed use the ideal mixing rules in order to combine the raw material properties with the blend ratios. In this paper, we present an alternative multi-block mixture PLS modeling approach that does not require the assumption of mixing rules. Furthermore, the multi-block

mixture PLS models are extended to the nonlinear modeling for the case where the ideal mixing rules are severely violated.

1.1. Data structure and problems

The databases generally available for mixture modeling problems are depicted in Fig. 1 for the case of blends of one class of raw materials (see Muteki et al. [1] for the structure when one has several classes of raw materials). The **R** matrix refers to the ($M \times N$) ratios (fractions) of all the materials used in the formation of blends. M is the number of blends, and N is the number of all the raw materials used in the blends. The data in **R** contains the fraction of each raw material used in the blend ($0 \leq r_{i,j} \leq 1.0$, and $\sum_{j=1}^N r_{i,j} = 1.0$). The ($M \times J$) **Z** matrix contains J process conditions used during manufacture of the blends, and the ($M \times L$) **Y** matrix contains L properties measured on the final product. The ($K \times N$) **X**^T matrix consists of the raw

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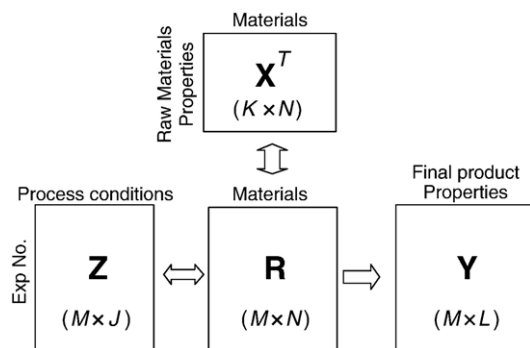


Fig. 1. T-shaped data structure generally available in mixture modeling.

material property data that corresponds to the subset of materials used in the available experimental blends $\mathbf{R}$. K is the number of raw material properties.

An important feature of this data structure is that there is no common dimension over all the data matrices. The $\mathbf{Z}$, $\mathbf{R}$ and $\mathbf{Y}$ matrices have one dimension in common (that corresponds to the experiment number or manufacturing conditions for making each final product). $\mathbf{X}^T$ has one dimension in common with the $\mathbf{R}$ matrix, but no dimension common to $\mathbf{Y}$. That is, $\mathbf{X}^T$ has an indirect relationship to $\mathbf{Y}$, only through $\mathbf{R}$. This kind of L-shape or T-shape data structure can occur in situations when auxiliary data such as $\mathbf{X}^T$ are obtained from a *different source*. The objective of mixture modeling is how to build latent variable models relating the $\mathbf{Z}$, $\mathbf{R}$ and $\mathbf{X}^T$ matrices to $\mathbf{Y}$ matrices. Of particular interest to researchers is the relationship between the raw material properties ($\mathbf{X}$) and final product properties ($\mathbf{Y}$). This information is crucial for the development of new products. A difficulty occurs because these matrices have no common dimensions. As a method to tackle this problem, the use of ideal mixing rules for combining the raw material properties ($\mathbf{X}$) and the blend ratios ($\mathbf{R}$) makes it possible to reduce the block matrices $\mathbf{X}$, $\mathbf{R}$, $\mathbf{Z}$ and $\mathbf{Y}$ with no common dimension to the matrices $\mathbf{X}_{\text{mix}}$ ($=\mathbf{R}\cdot\mathbf{X}$), $\mathbf{Z}$ and $\mathbf{Y}$ having a common dimension M [1,2]. A PLS model can then be used to relate these latter matrices. However, this approach needs the assumption that the ideal mixing rules are valid. This paper develops a more flexible mixture modeling approach that does not need such an assumption.

1.2. Review of multi-block approaches

Multi-block and hierarchical PCA and PLS methods have been proposed and are well discussed in the literature [3–6]. The advantage of these multi-block models is that they simplify the interpretability of the multivariate models. They have been used in cases where the number of variables is large and additional information is available for blocking the variables into conceptually meaningful blocks. Note that the nature of multi-blocks studied in most of these papers has a dimension in common (e.g. process time direction or molecule sample direction) with several data blocks in series (e.g. $\mathbf{Z}$, $\mathbf{R}$ and $\mathbf{Y}$ matrices in Fig. 1). Therefore, this problem is in essence different from the above mixture modeling problems in Fig. 1. Wold et al. [3] presented “two way PLS2” approaches for cases where the data blocks have

auxiliary objects. The advantage is that the multi-block approach can stabilize the model (i.e. loadings) by utilizing auxiliary data blocks consisting of additional X or Y data that is available but for which there are no corresponding X or Y data, respectively. However, for this approach the auxiliary data blocks must have the same set of variables (columns) as the common data blocks (X , Y), and this problem can be therefore regarded as the extension of the ordinary multi-block problems having one dimension in common.

Martens et al. [7] presented “L-PLSR”, a latent variable modeling approach for the L-shape data structure that occurs in food marketing data analysis problems. The L-shape data structure does not have a dimension in common, and is similar in structure to that of the L-shape mixture modeling treated here. However, it is quite different, in terms of modeling objectives and information flow, from the problems discussed in this paper. In particular, the objective of the L-PLSR problem [7] is to relate two different types of information matrices ($\mathbf{Z}$, $\mathbf{X}^T$) to the central matrix ($\mathbf{R}$) having one dimension in common with each of them (Fig. 1). In our case, the problem is to relate all of ($\mathbf{Z}$, $\mathbf{R}$, $\mathbf{X}^T$) to $\mathbf{Y}$ matrix in Fig. 1.

The paper is organized as follows. In Section 2, the mixture PLS models using ideal mixing rules are first briefly described and the new multi-block PLS modeling approach is presented. In Section 3, an industrial coke manufacturing example is used to demonstrate the effectiveness of the new multi-block approach. In Section 4, two nonlinear, multi-block, mixture PLS models are presented for cases where the ideal mixing rules are severely violated and/or other nonlinearities are present. Their effectiveness is illustrated by means of simulations where the nonlinearities are introduced in different ways.

2. L-shaped mixture PLS models

In this section, the mixture PLS models using ideal mixing rules [1,2] are first briefly described, and the new multi-block mixture PLS models are presented.

2.1. Mixture PLS models using ideal mixing rules

The traditional mixture models such as the Scheffé and Cox models [8], and mixture models based on PLS regression [9] are aimed at modeling only the relationship between the blend ratios ($\mathbf{R}$), and possibly process conditions ($\mathbf{Z}$) and blend properties ($\mathbf{Y}$). The model forms are generically expressed as

$$\mathbf{Y} = f(\mathbf{R}) + \varepsilon \quad (1)$$

Muteki et al. [1] presented a combined mixture-property modeling approach based on ideal mixing rules that allows for the incorporation of information on the raw material properties ($\mathbf{X}$) as well. The model is expressed as

$$\mathbf{Y} = f(\mathbf{X}_{\text{mix}}) + \varepsilon \quad (2)$$

where $\mathbf{X}_{\text{mix}} = \mathbf{R}\cdot\mathbf{X}$. The mixture PLS model of Eq. (2) represents the relationship between the ideal blend rule raw material properties of the mixture ($\mathbf{X}_{\text{mix}}$) and the final product properties ($\mathbf{Y}$). Combining the raw material properties ($\mathbf{X}$) and their mixture ratios ($\mathbf{R}$) through the use of mixing rules makes it possible to

reduce the block matrices $\mathbf{X}$, $\mathbf{R}$ and $\mathbf{Y}$ with no common dimension to the matrices $\mathbf{X}_{\text{mix}}$ and $\mathbf{Y}$ having a common dimension M . A Partial Least Square (PLS) regression model is then used to obtain the relationship in Eq. (2). Caution must be taken for the mean centering and scaling on each matrix. When calculating the ideal mixed properties $\mathbf{X}_{\text{mix}} = \mathbf{R} \cdot \mathbf{X}$, the raw data of both the blend ratios ($\mathbf{R}$) and the raw material properties ($\mathbf{X}$) are used (NOT the mean centered and scaled matrices). After that, $\mathbf{X}_{\text{mix}}$ and $\mathbf{Y}$ are mean centered and scaled to unit variance prior to PLS regression.

If the process operating conditions ($\mathbf{Z}$) change between blending experiments, then the effect of these changes are easily accounted for by incorporating $\mathbf{Z}$ into the PLS model as follows:

$$\mathbf{Y} = f(\mathbf{X}_{\text{mix}}, \mathbf{Z}) + \varepsilon \quad (3)$$

The mixture-property PLS models using ideal mixing rules need the following two assumptions. The first is that the ideal mixing rules are approximately valid. The second is that the available raw material properties correlate well with the final product properties ($\mathbf{Y}$). This latter condition will be naturally satisfied in most industrial applications because the raw material properties ($\mathbf{X}$) are measured because of their expected importance in any blending problems. Muteki [10] demonstrates that the new mixture-property PLS models work well in practical industrial examples, unless one of the above two conditions is severely violated. There are several advantages in this modeling approach that uses both $\mathbf{R}$ and $\mathbf{X}^T$, compared to the traditional mixture PLS model that use only $\mathbf{R}$: (1) it provides a direct interpretation of the effect of raw material properties ($\mathbf{X}$) on final product properties ($\mathbf{Y}$), that is, it provides information on *what raw material properties affect what mixture properties*, something that traditional mixture models using only the $\mathbf{R}$ and $\mathbf{Y}$ matrices cannot do; (2) the mixture-property PLS models provide better estimates of the blend properties ($\mathbf{Y}$) than the traditional mixture models in most industrial cases because of the use of the additional property information; (3) with new material grades which have never been used in the past, the mixture-property PLS models can estimate final product properties ($\mathbf{Y}$) given the ratios ($\mathbf{R}$) and the new raw material properties; (4) introducing the $\mathbf{X}_{\text{mix}}$ ($=\mathbf{R} \cdot \mathbf{X}$) matrix (i.e. ideal mixing rule) allows for the new approaches to the design of mixture experiments (DOE) that can simultaneously take into account both the effect of raw material properties and the effect of their blend ratios [2]; and (5) the mixture-property PLS models allows for the simultaneous selection of raw materials and their ratios in an optimization framework [1]. The same advantages will also hold for the multi-block L-shape mixture models proposed below.

2.2. Multi-block L-shaped mixture PLS models

For simplicity, the multi-block approach only for L-shape data structures consisting of $\mathbf{X}^T$, $\mathbf{R}$ and $\mathbf{Y}$ is described in this paper. If changing process conditions ($\mathbf{Z}$) are also present, the resulting T-shape data structure consisting of $\mathbf{X}^T$, $\mathbf{Z}$, $\mathbf{R}$ and $\mathbf{Y}$ can be treated by combining this new approach and the known multi-block PLS models [3–6] that can deal with the multiple data blocks with a common dimension.

Given the raw material property matrix $\mathbf{X}$, the blend ratio matrix $\mathbf{R}$ and the final blend property matrix $\mathbf{Y}$, the question is how to obtain a latent variable model with components (loadings and scores) shown in Fig. 2 to provide the best estimates on the final blend properties ($\mathbf{Y}$).

Prior to showing the modeling algorithm, the pre-processing is briefly described. Since the objective is to build mixture models for the final product properties, $\mathbf{Y}$ should be mean centered and scaled to unit variance for each of its L columns. Furthermore, since the models should be based on the raw material properties, $\mathbf{X}^T$ should be mean centered and scaled to unit variance for each of its N rows (raw material property direction). For $\mathbf{R}$, there can be three possible mean centering and scaling directions: (1) over the M rows; (2) over the N columns; (3) double mean centering [7]. In this study, $\mathbf{R}$ is mean centered and scaled to unit variance for each of the N columns. This centering and scaling approach for $\mathbf{R}$ was used because: (1) it corresponds to what is normally used in the absence of the auxiliary data matrix $\mathbf{X}$, and (2) it has provided the best estimates in the industrial examples studied by authors. However, the questions around mean centering and scaling the central matrix $\mathbf{R}$ in such LPLS algorithms are not straightforward (e.g. [7]).

After the mean centering and unit variance scaling on $\mathbf{X}$, $\mathbf{R}$ and $\mathbf{Y}$, a latent variable model for the first dimension can be represented as

$$\mathbf{X} = \mathbf{r} \cdot \mathbf{v}^T + \mathbf{E}_X \quad (4)$$

$$\mathbf{R} = \mathbf{t} \cdot \mathbf{p}^T + \mathbf{E}_R \quad (5)$$

$$\mathbf{Y} = \mathbf{u} \cdot \mathbf{q}^T + \mathbf{E}_Y \quad (6)$$

where $\mathbf{s}$ and $\mathbf{r}$ are the vectors of loading weights and scores of $\mathbf{X}$, $\mathbf{h}$ and $\mathbf{t}$ are the latent variable vectors for $\mathbf{R}$, $\mathbf{q}$ and $\mathbf{u}$ are the vectors of loading weights and scores for $\mathbf{Y}$. The loading vectors, $\mathbf{v}$ and $\mathbf{p}$, are computed after obtaining values of the above loading weight and score vectors at each component and used to deflate $\mathbf{X}$ and $\mathbf{R}$, respectively, to remove the information in these matrices that has been utilized in the current component. $\mathbf{E}_X$, $\mathbf{E}_R$ and $\mathbf{E}_Y$ represent the residuals.

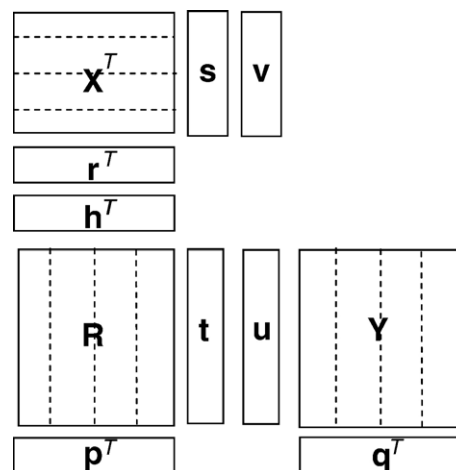


Fig. 2. Loadings and scores for multi-block mixture PLS models. (The dashed lines in each matrix indicate the mean centering and scaling direction.)

The latent vectors ($\mathbf{s}$, $\mathbf{r}$, $\mathbf{h}$, $\mathbf{t}$, $\mathbf{q}$ and $\mathbf{u}$) can be obtained iteratively using the modified NIPALS algorithm given in Appendix I. It is also shown in Appendix I that at convergence the modified NIPALS algorithm yields the loading weight vector $\mathbf{s}$ at each stage as the largest eigenvector of the matrix $\mathbf{X}^T \mathbf{R}^T \mathbf{Y} \mathbf{Y}^T \mathbf{R} \mathbf{X}$. Following Burnham et al. [11] this is equivalent to using the following objective function on the deflated matrices at each stage.

$$\max_{\mathbf{s}} \mathbf{s}^T \mathbf{X}^T \mathbf{R}^T \mathbf{Y} \mathbf{Y}^T \mathbf{R} \mathbf{X} \mathbf{s} \quad (7)$$

$$\text{s.t. } \mathbf{s}^T \mathbf{s} = 1$$

This result provides an interesting link between this multi-block L-shaped PLS regression approach and the previous approach using the ideal mixing rules to combine $\mathbf{X}^T$ and $\mathbf{R}$ into a single matrix ($\mathbf{X}_{\text{mix}} = \mathbf{R} \cdot \mathbf{X}$) followed by ordinary PLS. Clearly the objective function (Eq. (7)) shows that this solution to the multi-block L-shaped PLS problem is equivalent to performing an ordinary PLS regression on $\mathbf{R} \cdot \mathbf{X}$ vs. $\mathbf{Y}$. However, the equivalence is not exact because in this multi-block approach each matrix in Eq. (7) is separately mean centered and scaled to unit variance prior to the modeling, whereas the ideal mixing rules apply to the unscaled $\mathbf{X}^T$ and $\mathbf{R}$ matrices, and so in that approach the $\mathbf{R} \cdot \mathbf{X}$ matrix is first formed with no scaling and then $\mathbf{X}_{\text{mix}}$ ($= \mathbf{R} \cdot \mathbf{X}$) is scaled to unit variance prior to PLS regression. Aside from this scaling issue, the two approaches appear to solve a similar objective function.

Once the vector of loading weights $\mathbf{s}$ is computed by Eq. (7) or by the iterative NIPALS algorithm, the rest of the vectors can be computed as

$$\mathbf{r} = \mathbf{X} \mathbf{s} (\mathbf{s}^T \mathbf{s})^{-1} \quad (8)$$

$$\mathbf{t} = \mathbf{R} \mathbf{r} / (\mathbf{r}^T \mathbf{r})^{-1} \quad (9)$$

$$\mathbf{q} = \mathbf{Y}^T \mathbf{t} / (\mathbf{t}^T \mathbf{t})^{-1} \quad (10)$$

$$\mathbf{u} = \mathbf{Y} \mathbf{q} / (\mathbf{q}^T \mathbf{q})^{-1} \quad (11)$$

$$\mathbf{h} = \mathbf{R}^T \mathbf{u} / (\mathbf{u}^T \mathbf{u})^{-1} \quad (12)$$

In order to deflate the matrices to remove the information explained by the first component, the loading vectors $\mathbf{p}$ and $\mathbf{v}$ can be computed as

$$\mathbf{p} = \mathbf{R}^T \mathbf{t} / (\mathbf{t}^T \mathbf{t})^{-1} \quad (13)$$

$$\mathbf{v} = \mathbf{X}^T \mathbf{r} / (\mathbf{r}^T \mathbf{r})^{-1} \quad (14)$$

Before computing the vectors of loading weights for the next component, the data matrices are deflated as

$$\mathbf{X} = \mathbf{X} - \mathbf{r} \cdot \mathbf{v}^T \quad (15)$$

$$\mathbf{R} = \mathbf{R} - \mathbf{t} \cdot \mathbf{p}^T \quad (16)$$

$$\mathbf{Y} = \mathbf{Y} - \mathbf{t} \cdot \mathbf{q}^T \quad (17)$$

By deflating the matrix $\mathbf{R}$ by $\mathbf{t} \cdot \mathbf{p}^T$ (rather than $\mathbf{t} \cdot \mathbf{h}^T$) and $\mathbf{X}$ by $\mathbf{r} \cdot \mathbf{v}^T$ (rather than $\mathbf{r} \cdot \mathbf{s}^T$) one is removing only information in $\mathbf{R}$ and $\mathbf{X}$ that was actually utilized in computing the first

dimension score $\mathbf{t}$. This issue is well discussed in the multi-block modeling literature [5,6,12].

Normalizing alternative latent vectors, other than $\mathbf{s}$, would simply lead to alternative eigenvalue–eigenvector solutions, and the other loading and score vectors would again be determined as in Eqs. (8)–(12).

There are several features and advantages in the multi-block mixture PLS models: (1) the multi-block mixture-property PLS models can provide better interpretability than the mixture-property PLS models using ideal mixing rules, since the latent components $\mathbf{r}$ and $\mathbf{h}$ corresponding to the materials are available for interpretation. This allows one a more direct interpretation of the relationship between the raw material properties and the final product properties. This will be illustrated in the industrial example in Section 3; (2) the iterative NIPALS algorithm can be adapted in the usual manner to easily handle missing data in the $\mathbf{X}$, $\mathbf{R}$ and $\mathbf{Y}$ matrices. An alternative approach for imputing missing data in $\mathbf{X}$ (usually the matrix with most missing data) using the information in $\mathbf{R}$ and $\mathbf{Y}$ as auxiliary data is given in Muteki et al. [13]; (3) the multi-block PLS model can easily be extended to incorporate nonlinearities in either or both of the relationships between the scores $\mathbf{t}$ and $\mathbf{u}$, or $\mathbf{r}$ and $\mathbf{h}$. These nonlinear multi-block models are presented and illustrated in Section 4. They allow for deviations for the blending rules, or for the modeling of unknown and complex blending problems.

On the other hand, building the multi-block mixture-property PLS models for the case of blends of multiple classes of raw materials [1,2] and for the case of T-shape data structures including the process conditions $\mathbf{Z}$ is more cumbersome as the eigenvector solution or the equivalent NIPALS algorithm will be more complex. For these cases, the mixture-property PLS model using mixing rules is very flexible and still easy to implement as shown in Eq. (3).

3. Industrial example

An industrial example, using the multi-block mixture PLS models, is presented in this section. The example features the production of cokes for use in steel making. Cokes are manufactured from mixtures of several different coals processed in coke ovens. In this example it is assumed that the coke property matrix $\mathbf{Y}$ is influenced by only the coal blend ratios ($\mathbf{R}$) and the properties of the coals ($\mathbf{X}^T$), since the same manufacturing equipment was used in all cases and the processing conditions of the coke oven were kept almost constant ($\mathbf{Z} = \text{constant}$ in Fig. 1). Therefore, this problem has an L-shape data structure as shown in Fig. 2.

In this industrial application, the (54×17) coal property matrix $\mathbf{X}^T$ consists of measurements of 54 properties on 17 coals, the (63×17) coal blend ratio matrix $\mathbf{R}$ consists of 63 blends and the ratios of the 17 coals used in each blend, and the (63×3) coke property matrix $\mathbf{Y}$ consists of 3 coke properties measured on each of the 63 final cokes. The explained variances using both the mixture-property PLS models based on ideal mixing rules and based on the multi-block mixture PLS models are shown in Table 1.

Both approaches give almost identical results. In product development problems or in this case the selection of coals that

Table 1
Explained variances on coke blend properties Y (12 PLS components)

	Explained Y (%)
(1) Mixture-property PLS model using ideal mixing rules	76.77
(2) Multi-block mixture-property PLS model	76.36

will yield cokes with desired properties, it is very important for researchers to develop a feel for variable relationships among the coal properties (**X**), the coal blend ratios (**R**) and coke properties (**Y**). A side of benefit of the multi-block PLS models is that it can be used to more directly provide qualitative insight into these relationships through the loading plots and the score plots as shown in Fig. 3(a–d).

The arrows in the plots indicate the directions that control the Y2 and Y3 coke properties (see Fig. 3(a)). The blend samples of each circle area drawn along these arrow directions in the t1–t2

score plot (in Fig. 3(b)) indicate coal blends that result in large and small Y2 and Y3 coke properties, respectively. For instance, the blend mixtures 42 and 49 lead to a very small Y2 property. The coals shown in each circle area along the arrow directions in the r1–r2 score plot (in Fig. 3(c)) indicate the dominant coals used in the blends of each circle area depicted in Fig. 3(b), respectively. For instance, the blend sample 42 consists of 100% blend of R13 coal. The most useful plot for researchers and product development personal is the final plot of the loadings (s1–s2) of the **X** matrix (Fig. 3(d)) which shows how the 54 coal raw material properties are related to the final coke properties. The coal properties in each circle area along the arrow directions in s1–s2 score plot (in Fig. 3(d)) are those that have most influence on the Y2 and Y3 properties of the cokes. This information illustrates one of the advantages of this L-shaped PLS modeling; relationships between the final product properties (**Y**) and the raw material properties (**X**) can be

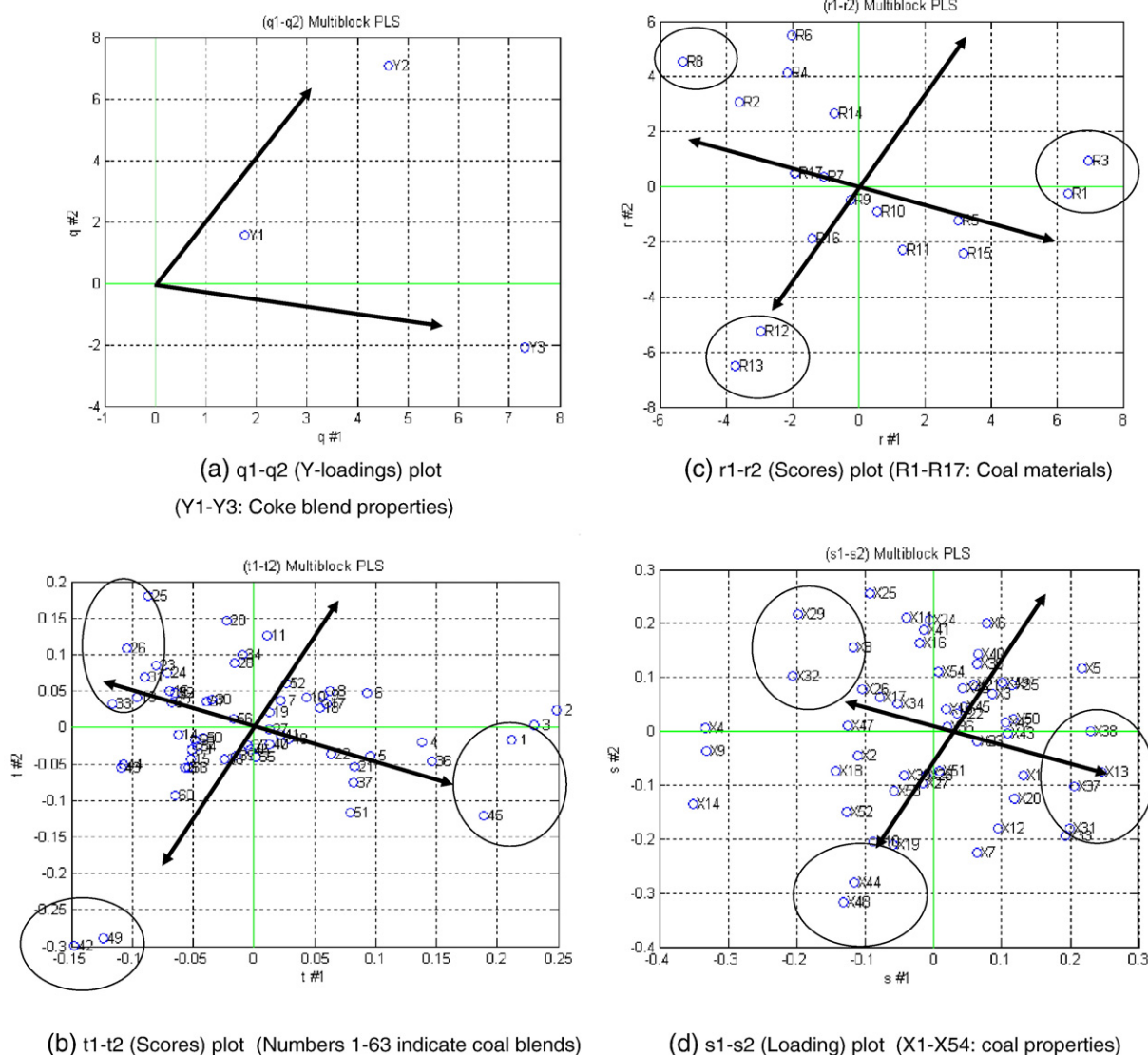


Fig. 3. (a) q1–q2 (Y-loadings) plot (Y1–Y3: coke blend properties). (b) t1–t2 (scores) plot (numbers 1–63 indicate coal blends). (c) r1–r2 (scores) plot (R1–R17: coal materials). (d) s1–s2 (loading) plot (X1–X54: coal properties).

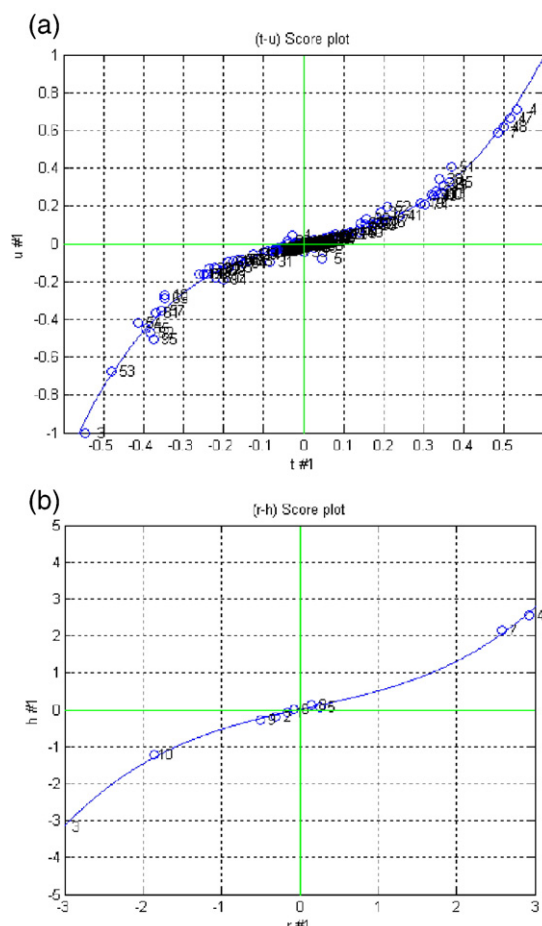


Fig. 4. (a) The first PLS component of scores $\mathbf{t}$ and $\mathbf{u}$ captured by the linear multi-block mixture PLS model in Example 1 (the line shows the estimates fitted by a cubic polynomial nonlinear function). (b) The first PLS component of scores $\mathbf{r}$ and $\mathbf{h}$ captured by the linear multi-block mixture PLS model in Example 1 (the line shows the estimates fitted by the cubic polynomial nonlinear function).

obtained in spite of the fact that these matrices have no common dimension.

4. Extension to nonlinear models

In this section the multi-block methods are extended to allow for various types of nonlinearities among the different blends. Two nonlinear multi-block PLS algorithms are presented. The nonlinear algorithm 1 takes into account the nonlinearity

between the $\mathbf{t}$ and $\mathbf{u}$ scores in Fig. 2, and the nonlinear algorithm 2 takes into account the nonlinearity between the $\mathbf{r}$ and $\mathbf{h}$ scores in Fig. 2. The detailed algorithms for these 2 cases are given in Appendix II. In this study a cubic polynomial function is specifically used for fitting the inner relationships (between $\mathbf{t}$ and $\mathbf{u}$, between $\mathbf{r}$ and $\mathbf{h}$), but these algorithms can be modified to include any nonlinear function.

4.1. Nonlinear mixture model simulations

Two types of nonlinearities are studied in this section. One is where the true raw material properties of the blends ($\mathbf{RX}_{\text{true}}$) are represented as nonlinear functions of the raw material properties of the blends based on ideal mixing rules, expressed in Eq. (19).

$$\mathbf{RX}_{\text{true}} = f(\mathbf{R} \cdot \mathbf{X}) \quad (19)$$

Since the final product properties $\mathbf{Y}$ should be related to $\mathbf{RX}_{\text{true}}$, it is expected that algorithm 1 that provides a nonlinear model between $\mathbf{t}$ and $\mathbf{u}$ should handle this situation the best. Another

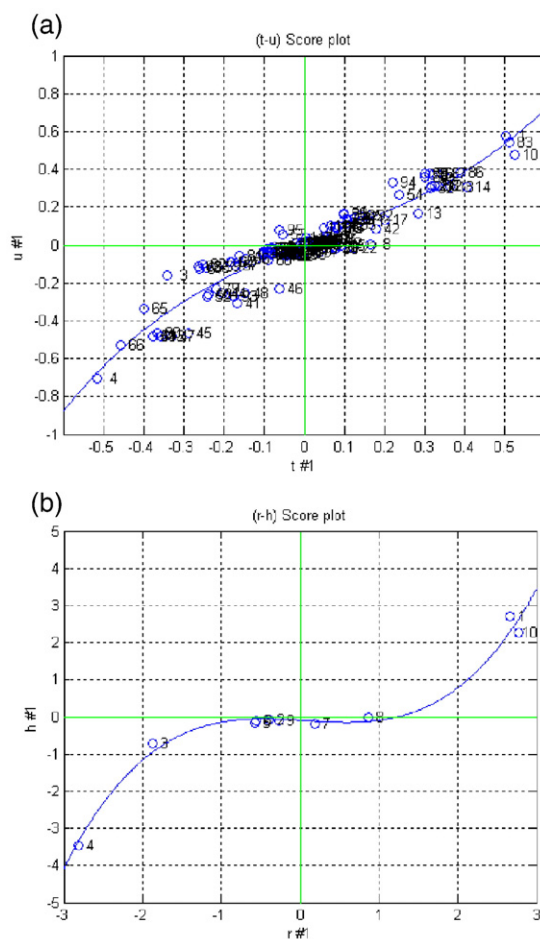


Fig. 5. (a) The first PLS component of scores $\mathbf{t}$ and $\mathbf{u}$ captured by the linear multi-block mixture PLS model in Example 2 (the line shows the estimates fitted by a cubic polynomial nonlinear function). (b) The first PLS component of scores $\mathbf{r}$ and $\mathbf{h}$ captured by the linear multi-block mixture PLS model in Example 2 (the line shows the estimates fitted by a cubic polynomial nonlinear function).

Table 2

Results on explained variance from each modeling approach (Example 1) based on the average of 100 Monte Carlo simulations

Method	Explained Y (%, averaged)
(1) Linear mixture PLS model using ideal mixing rules	88.2755
(2) Linear multi-block mixture PLS model	88.2649
(3) Nonlinear algorithm 1	98.6432
(4) Nonlinear algorithm 2	90.1096

Table 3
Results on explained variance from each modeling approach (Example 2) based on the average of 100 Monte Carlo simulations

Method	Explained Y (%, averaged)
(1) Linear mixture PLS model using ideal mixing rules	84.6968
(2) Linear multi-block mixture PLS model	84.6144
(3) Nonlinear algorithm 1	87.8283
(4) Nonlinear algorithm 2	99.7741

situation is where the true raw material properties of the blends ($\mathbf{RX}_{\text{true}}$) arises from nonlinearities in the raw material properties affecting the blend properties $\mathbf{RX}_{\text{true}}$, expressed in Eq. (20).

$$\mathbf{RX}_{\text{true}} = \mathbf{R} \cdot f(\mathbf{X}) \quad (20)$$

In this case the nonlinearities occur mainly between $\mathbf{R}$ and $\mathbf{X}$ matrix and algorithm 2 (nonlinear relationship between $\mathbf{r}$ and $\mathbf{h}$) might be expected to provide the best model.

Example 1. In this first simulation, we consider the nonlinearity in Eq. (19) ($\mathbf{RX}_{\text{true}} = f(\mathbf{R} \cdot \mathbf{X})$) where the raw material property matrix $\mathbf{X}$ has two directions and the product property $\mathbf{Y}$ involves both a linear and a nonlinear term in the ideal mixing term ($\mathbf{R} \cdot \mathbf{X}$).

$$\mathbf{X} = [\mathbf{d1} \ \mathbf{d1} \ \mathbf{d2} \ \mathbf{d2} \ \mathbf{d2}]$$

$$\mathbf{Y} = \mathbf{R} \cdot \mathbf{X} + (\mathbf{R} \cdot \mathbf{X})^3 + \mathbf{E}$$

where $\mathbf{d1}$ and $\mathbf{d2}$ are the random vectors of unit length and they are orthogonal to each other, $\mathbf{E}$ is an error matrix. The (10×5) raw material property matrix $\mathbf{X}$ contains 5 properties on the 10 raw materials, the (133×10) blend ratio matrix $\mathbf{R}$ contains the ratios of the 10 raw materials used in 133 blends, and the (133×5) blend property matrix $\mathbf{Y}$ contains 5 final product properties obtained from each of the 133 blends. In this study the 133 entries in the blend ratio matrix $\mathbf{R}$ were obtained by a D-optimal design, and hence $\mathbf{R}$ is well conditioned. A Monte Carlo simulation study was performed in which 100 simulations of the model were performed using different random error $\mathbf{E}$ and different random vectors $\mathbf{d1}$ and $\mathbf{d2}$. The blend ratio matrix $\mathbf{R}$ was the same for all 100 Monte Carlo simulations.

In order to illustrate the nonlinearities, the score plots of the first PLS components captured by one of the linear multi-block PLS models are shown in Fig. 4 ((a) scores $\mathbf{t}$ vs. $\mathbf{u}$, (b) scores $\mathbf{r}$ vs. $\mathbf{h}$).

From the nonlinearities in Fig. 4, it is expected that the estimates can be improved by using the nonlinear models fitting the inner relations. Results of the explained variance obtained from different approaches are shown in Table 2.

The results of the two linear mixture PLS models are almost equivalent as expected from the discussion in Section 2.2. As expected, the nonlinear algorithm 1 that takes into account the nonlinearity between $\mathbf{t}$ and $\mathbf{u}$ scores provides the best estimates, because the nonlinearities in this case are mainly between $\mathbf{Y}$ and $\mathbf{RX}_{\text{true}}$ ($=\mathbf{R} \cdot \mathbf{X} + (\mathbf{R} \cdot \mathbf{X})^3$). The nonlinear

algorithm 2 could not improve the estimates much, since the assumed nonlinearity does not appear between $\mathbf{X}$ and $\mathbf{R}$. Clearly the effectiveness of the approaches depends on where the nonlinearity appears.

Example 2. In a second simulation, we consider the nonlinearity in Eq. (20) ($\mathbf{RX}_{\text{true}} = \mathbf{R} \cdot f(\mathbf{X})$) where the raw material property matrix $\mathbf{X}$ again has two directions ($\mathbf{d1}$ and $\mathbf{d2}$ are orthogonal random vectors) but the mixing rule for the blend properties is assumed to hold for the cubic values of the material properties, that is

$$\mathbf{X} = [\mathbf{d1} \ \mathbf{d1} \ \mathbf{d2} \ \mathbf{d2} \ \mathbf{d2}]$$

$$\mathbf{Y} = \mathbf{R} \cdot \mathbf{X}^3 + \mathbf{E}$$

where $\mathbf{X}^3 = [\mathbf{d1}^3 \ \mathbf{d1}^3 \ \mathbf{d2}^3 \ \mathbf{d2}^3 \ \mathbf{d2}^3]$. The data size of each matrix is the same as in Example 1 and the Monte Carlo simulations were also performed as in Example 1. In order to illustrate the nonlinearities, the score plots of the first PLS components captured by one of the linear multi-block PLS models are shown in Fig. 5 ((a) scores $\mathbf{t}$ vs. $\mathbf{u}$, (b) scores $\mathbf{r}$ vs. $\mathbf{h}$).

The results on the explained variance from each modeling approach are shown in Table 3.

Once again, the results of the two linear mixture PLS models are almost equivalent as expected. In this simulation, the nonlinear algorithm 2 that takes into account the nonlinearity between the scores $\mathbf{r}$ and $\mathbf{h}$ provides the best estimates since the simulated nonlinearity mainly appears between the $\mathbf{X}$ and $\mathbf{R}$ matrices (non-ideal mixing).

5. Conclusion

This paper presents new multi-block mixture-property PLS models that are well suited for mixture-property modeling. Linear multi-block PLS regression models are theoretically shown to have a close relationship to the linear PLS model based on ideal mixing rules and, as a result, the two approaches give nearly identical results for both the industrial data and the simulation studies. However, the loading vectors of the multi-block approach are shown to provide important additional insights into the relationships between the raw material properties and the final product properties. Furthermore, the multi-block methods are easily extended to account for nonlinearities in either direction of the L-shaped data structure, namely between the $\mathbf{X}$ and $\mathbf{R}$ (non-ideal mixing) or between the true blend properties ($\mathbf{RX}_{\text{true}}$) and the product properties $\mathbf{Y}$. The performance of nonlinear multi-block PLS methods is illustrated for the two non-ideal situations where ideal mixing rules do not hold.

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Appendix A. Modified NIPALS algorithm for L-shaped multi-block PLS, and an equivalent eigenvalue–eigenvector solution

The modified NIPALS algorithm is:

1. Initialize $\mathbf{u}$ with the first column of $\mathbf{Y}$
2. $\mathbf{h} = \mathbf{R}^T \mathbf{u} (\mathbf{u}^T \mathbf{u})^{-1}$
3. $\mathbf{s} = \mathbf{X}^T \mathbf{h} (\mathbf{h}^T \mathbf{h})^{-1}$
4. Normalize $\mathbf{s}$ as $\|\mathbf{s}\| = 1$
5. $\mathbf{r} = \mathbf{X} \mathbf{s} (\mathbf{s}^T \mathbf{s})^{-1}$
6. $\mathbf{t} = \mathbf{R} \mathbf{r} (\mathbf{r}^T \mathbf{r})^{-1}$
7. $\mathbf{q} = \mathbf{Y}^T \mathbf{t} (\mathbf{t}^T \mathbf{t})^{-1}$
8. $\mathbf{u} = \mathbf{Y} \mathbf{q} (\mathbf{q}^T \mathbf{q})^{-1}$
9. Convergence is checked with respect to $\mathbf{u}$. If convergence fails, use new value of $\mathbf{u}$ and go to Step 2
10. $\mathbf{p} = \mathbf{R}^T \mathbf{t} (\mathbf{t}^T \mathbf{t})^{-1}$
11. Deflate $\mathbf{X}$, $\mathbf{R}$ and $\mathbf{Y}$ as

$$\mathbf{X} = \mathbf{X} - \mathbf{r} \cdot \mathbf{v}^T$$

$$\mathbf{R} = \mathbf{R} - \mathbf{t} \cdot \mathbf{p}^T$$

$$\mathbf{Y} = \mathbf{Y} - \mathbf{u} \cdot \mathbf{q}^T$$

At convergence of the modified NIPALS, using the regression equations in Steps 2–8, the weight vector $\mathbf{s}$ is given as

$$\mathbf{s} = \mathbf{X}^T \mathbf{R}^T \mathbf{Y} \mathbf{Y}^T \mathbf{R} \mathbf{X} \mathbf{s} (\mathbf{h}^T \mathbf{h})^{-1} (\mathbf{u}^T \mathbf{u})^{-1} \times (\mathbf{q}^T \mathbf{q})^{-1} (\mathbf{t}^T \mathbf{t})^{-1} (\mathbf{r}^T \mathbf{r})^{-1} (\mathbf{s}^T \mathbf{s})^{-1}$$

Setting $\rho^{-1} = (\mathbf{h}^T \mathbf{h})^{-1} (\mathbf{u}^T \mathbf{u})^{-1} (\mathbf{q}^T \mathbf{q})^{-1} (\mathbf{t}^T \mathbf{t})^{-1} (\mathbf{r}^T \mathbf{r})^{-1} (\mathbf{s}^T \mathbf{s})^{-1}$, the weight vector $\mathbf{s}$ is rewritten as

$$\mathbf{X}^T \mathbf{R}^T \mathbf{Y} \mathbf{Y}^T \mathbf{R} \mathbf{X} \mathbf{s} = \rho \mathbf{s}$$

This expression shows that NIPALS algorithm corresponds to the eigenvalue–eigenvector solution where $\mathbf{s}$ is the eigenvector of $\mathbf{X}^T \mathbf{R}^T \mathbf{Y} \mathbf{Y}^T \mathbf{R} \mathbf{X}$.

Appendix B. Nonlinear multi-block PLS algorithms

This algorithm can be modified to include any nonlinear function between the scores ($\mathbf{t}$ and $\mathbf{u}$, $\mathbf{r}$ and $\mathbf{h}$). The present algorithm is specifically for the case of a cubic polynomial function. The algorithms are similar to the one for the nonlinear PLS methods proposed by Wold et al. [14,15].

Algorithm 1. Nonlinearity between $\mathbf{t}$ and $\mathbf{u}$

1. $a=1$ (a denotes component)
2. Initialize $\mathbf{u}_a$ and $\mathbf{t}_a$ with the linear multi-block PLS algorithm
3. Fit b_{0a} , b_{1a} , b_{2a} and b_{3a} with the initial $\mathbf{u}_a$ and $\mathbf{t}_a$ based on the cubic polynomial function $\mathbf{u}_a = b_{0a} + b_{1a} \mathbf{t}_a + b_{2a} \mathbf{t}_a^2 + b_{3a} \mathbf{t}_a^3$ using ordinary least squares.

4. Calculate $\hat{\mathbf{u}}_a$ using the fitted cubic in Step 3.

$$5. \mathbf{q}_a^T = \hat{\mathbf{u}}_a^T \cdot \mathbf{Y} / (\hat{\mathbf{u}}_a^T \hat{\mathbf{u}}_a)$$

$$6. \mathbf{u}_a^T = \mathbf{Y} \mathbf{q}_a / (\mathbf{q}_a^T \mathbf{q}_a)$$

7. Update $\mathbf{s}_a$ by nonlinear regression such that

$$\mathbf{u}_a = b_{0a} + b_{1a} \mathbf{t}_a + b_{2a} \mathbf{t}_a^2 + b_{3a} \mathbf{t}_a^3$$

$$\text{where } \mathbf{t}_a^T = \mathbf{r}_a^T \mathbf{R} / (\mathbf{r}_a^T \mathbf{r}_a) \text{ and } \mathbf{r}_a = \mathbf{X} \mathbf{s}_a$$

8. Normalize $\mathbf{s}_a$ as $\|\mathbf{s}_a\| = 1$
9. Check convergence in $\mathbf{t}_a$

10. If not converged return to Step 3

11. If converged, calculate $\mathbf{v}_a$ as $\mathbf{v}_a = \mathbf{X}_a^T \mathbf{r}_a / (\mathbf{r}_a^T \mathbf{r}_a)$ and $\mathbf{p}_a$ as $\mathbf{p}_a = \mathbf{R}_a^T \mathbf{t}_a / (\mathbf{t}_a^T \mathbf{t}_a)$

12. Deflate $\mathbf{X} = \mathbf{X} - \mathbf{r}_a \cdot \mathbf{v}_a^T$, $\mathbf{R} = \mathbf{R} - \mathbf{t}_a \cdot \mathbf{p}_a^T$ and $\mathbf{Y} = \mathbf{Y} - \mathbf{u}_a \cdot \mathbf{q}_a^T$

13. $a = a + 1$, go to Step 2.

Algorithm 2. Nonlinearity between $\mathbf{r}$ and $\mathbf{h}$

1. $a=1$ (a denotes component)

2. Initialize $\mathbf{h}_a$ and $\mathbf{r}_a$ with the linear multi-block PLS algorithm

3. Fit c_{0a} , c_{1a} , c_{2a} and c_{3a} with the initial $\mathbf{h}_a$ and $\mathbf{r}_a$ based on the cubic polynomial function $\mathbf{h}_a = c_{0a} + c_{1a} \mathbf{r}_a + c_{2a} \mathbf{r}_a^2 + c_{3a} \mathbf{r}_a^3$ using ordinary least squares.

4. Calculate $\hat{\mathbf{h}}_a$ using the fitted cubic in Step 3.

$$5. \mathbf{t}_a^T = \mathbf{R} \hat{\mathbf{h}}_a^T / (\hat{\mathbf{h}}_a^T \hat{\mathbf{h}}_a)$$

$$6. \mathbf{h}_a^T = \mathbf{R}^T \mathbf{t}_a / (\mathbf{t}_a^T \mathbf{t}_a)$$

7. Update $\mathbf{s}_a$ by nonlinear regression such that

$$\mathbf{h}_a = c_{0a} + c_{1a} \mathbf{r}_a + c_{2a} \mathbf{r}_a^2 + c_{3a} \mathbf{r}_a^3$$

$$\text{where } \mathbf{r}_a^T = \mathbf{X} \mathbf{s}_a / (\mathbf{s}_a^T \mathbf{s}_a)$$

8. Normalize $\mathbf{s}_a$ as $\|\mathbf{s}_a\| = 1$
9. $\mathbf{q}_a = \mathbf{Y}^T \mathbf{t}_a / (\mathbf{t}_a^T \mathbf{t}_a)$

$$10. \mathbf{u}_a = \mathbf{Y} \mathbf{q}_a / (\mathbf{q}_a^T \mathbf{q}_a)$$

11. Check convergence in $\mathbf{t}_a$

12. If not converged return to Step 3

13. If converged, calculate $\mathbf{v}_a$ as $\mathbf{v}_a = \mathbf{X}_a^T \mathbf{r}_a / (\mathbf{r}_a^T \mathbf{r}_a)$ and $\mathbf{p}_a$ as $\mathbf{p}_a = \mathbf{R}_a^T \mathbf{t}_a / (\mathbf{t}_a^T \mathbf{t}_a)$

14. Deflate $\mathbf{X} = \mathbf{X} - \mathbf{r}_a \cdot \mathbf{v}_a^T$, $\mathbf{R} = \mathbf{R} - \mathbf{t}_a \cdot \mathbf{p}_a^T$ and $\mathbf{Y} = \mathbf{Y} - \mathbf{u}_a \cdot \mathbf{q}_a^T$

15. $a = a + 1$, go to Step 2.

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